

Syllabus for R-SAT

Section 1: English

Vocabulary: Synonyms and Antonyms, Meaning of Words in Context, Idioms and Phrases

Grammar: Error Detection, Sentence Correction, Appropriate Use of Prepositions, Articles, and Tenses

Sentence Construction: Sentence Formation, Correctly Punctuated Sentences

Comprehension Skills: Fill-in-the-Blanks, Understanding Word Usage

Section 2: Logical Reasoning

Verbal Reasoning: Relationships (e.g., Family Trees, Statements), Direction Sense

Series and Patterns: Number and Letter Series, Missing Numbers/Letters

Puzzles: Logical Puzzles, Symbol-Based Reasoning

Analogy and Classification: Identifying Differences and Similarities

Critical Thinking: Evaluating Logical Consistency.

Section 3: Aptitude/Mathematics

Arithmetic: Profit and Loss, Percentages and Ratio, Simple and Compound Interest

Geometry and Measurement: Area, Volume, and Perimeter, Heights and Distances

Algebra: Linear Equations, Basic Algebraic Expressions

Statistics and Averages: Mean, Median, and Mode

Number Systems: Prime Numbers, Factors and Multiples, Basic Operations

(Addition, Subtraction, Multiplication, Division), Number System Conversions (e.g., binary to decimal, octal, hexadecimal, and vice versa).

Time and Distance: Speed, Distance, and Time, Work and Efficiency

Section 4: Computer Basic Knowledge

Computer Hardware and Software: Basic Components of a Computer, Functions of Input and Output Devices

Operating Systems: Roles and Features of the Kernel, File Systems and File Management

Memory and Storage: Types of Memory (RAM, ROM, Cache), Data Storage Devices (HDD, SSD)

Networking Basics: IP Addressing, Basic Network Components, mac address

Computer Terminology: Common Acronyms (e.g., URL, HTTP, ACID), Functionalities of Frequently Used Keys (e.g., F5 for refresh)

Basic Troubleshooting: Using Task Manager, Managing Deleted Files (Recycle Bin)

Security Basics: Understanding the Role of Firewalls, Basic Cybersecurity Practices

SAMPLE R-SAT Paper

English Section

1. Choose the correct meaning of the underlined word. The scientist gave a comprehensive explanation of the experiment.

- a) Detailed
- b) Short
- c) Confusing
- d) Incomplete

Correct Answer: a) Detailed

2. Choose the word which is opposite in meaning to the given word.

Generous

- a) Selfish
- b) Kind
- c) Rich
- d) Helpful

Correct Answer: a) Selfish

3. Fill in the blank. She is the most _____ student in our class.

- a) Intelligent
- b) Intelligence
- c) Intelligently
- d) More intelligent

Correct Answer: a) Intelligent

4. Choose the correct sentence.

- a) He don't like reading books.
- b) He doesn't like reading books.
- c) He didn't like reading books.
- d) He does not liked reading books.

Correct Answer: b) He doesn't like reading books.

5. Choose the correct word to complete the sentence. The team was _____ by the coach.

- a) Lead
- b) Led
- c) Leading
- d) Leaded

Correct Answer: b) Led

6. Identify the error in the following sentence. The team have won the match.

- a) The
- b) Have
- c) Won
- d) Match

Correct Answer: b) Have

7. Choose the synonym of the word 'Sincere'.

- a) Honest
- b) Pretentious
- c) Insincere
- d) Kind

Correct Answer: a) Honest

8. Fill in the blank with the correct form of the verb. She _____ her homework already.

- a) Finished
- b) Has finished
- c) Have finished
- d) Is finishing

Correct Answer: b) Has finished

9. **Choose the correct meaning of the idiom 'Break the ice'.**

- a) Destroy something
- b) Start a conversation
- c) Make someone cry
- d) Become frozen

Correct Answer: b) Start a conversation

10. **Choose the correct sentence.**

- a) I have been to London last year.
- b) I had been to London last year.
- c) I went to London last year.
- d) I go to London last year.

Correct Answer: c) I went to London last year.

11. **Fill in the blank. This is the book _____ I was talking about.**

- a) Who
- b) Whom
- c) Which
- d) Where

Correct Answer: c) Which

12. **Choose the correct preposition. She is interested _____ learning French.**

- a) In
- b) On
- c) At
- d) Of

Correct Answer: a) In

13. **Choose the correctly punctuated sentence:**

- a) "The professor said," read chapters five and six for tomorrow's lecture."
- b) "The professor said, "Read chapters five and six for tomorrow's lecture."
- c) The professor said "read chapters five and six for tomorrow's lecture."
- d) The professor said, "read chapters five and six for tomorrow's lecture."

Correct Answer: b) "The professor said, "Read chapters five and six for tomorrow's lecture."

14. **Choose the word that is closest in meaning to 'Benevolent'.**

- a) Malevolent
- b) Kind
- c) Rich
- d) Strong

Correct Answer: b) Kind

15. **Fill in the blank with the appropriate article. She is _____ best singer I know.**

- a) A
- b) An
- c) The
- d) None of the above

Correct Answer: c) The

Logical Reasoning Section

1. **If all roses are flowers and some flowers are red, can we say that some roses are red?**

- a) Yes
- b) No
- c) Not enough information
- d) Impossible

Correct Answer: c) Not enough information

2. Which number is missing in the sequence: 2, 6, 12, 20, ____?

- a) 30
- b) 28
- c) 25
- d) 24

Correct Answer: a) 30

3. If a clock shows the time as 3:15, what is the angle between the hour and minute hands?

- a) 45°
- b) 30°
- c) 7.5°
- d) 15°

Correct Answer: c) 7.5°

4. If $A + B$ means A is the brother of B; $A - B$ means A is the sister of B and $A \times B$ means A is the father of B. Which of the following means that C is the son of M?

- a) $M - N \times C + F$
- b) $F - C + N \times M$
- c) $N + M - F \times C$
- d) $M \times N - C + F$

Correct Answer: D) $M \times N - C + F$

5. Yard is to inch as quart is to

- a) gallon
- b) ounce
- c) milk
- d) liquid

Correct Answer: b) ounce

6. A man walks 5 km toward south and then turns to the right. After walking 3 km he turns to the left and walks 5 km. Now in which direction is he from the starting place?

- a) West
- b) South
- c) North-East
- d) South-West

Correct Answer: D) South-West

7. Which one of the following is always in 'Sentiment'?

- a) Cruelty
- b) Insight
- c) Neutrality
- d) Emotion

Correct Answer: d) Emotion

8. CMM, EOO, GQQ, _____, KUU

- a) GRR
- b) GSS
- c) ISS
- d) ITT

Correct Answer: C) ISS

9. Choose the word which is different from the rest.

- a) Kiwi
- b) Eagle
- c) Emu
- d) Ostrich

Correct Answer: b) Eagle

10. Look at this series: F2, __, D8, C16, B32, ... What number should fill the blank?

- a) A16
- b) G4
- c) E3
- d) E4

Correct Answer: d) E4

11. Look carefully at the sequence of symbols to find the pattern. Select correct pattern.



- a) 1
- b) 2
- c) 3
- d) 4

Correct Answer: c) 3

12. Look at this series: 8, 43, 11, 41, __, 39, 17, ... What number should fill in the blank?

- a) 8
- b) 14
- c) 43
- d) 44

Correct Answer: b) 14

13. **Sachin walks 20 km towards North. He turns left and walks 40 km. He again turns left and walks 20 km. Finally he moves 20 km after turning to the left. How far is he from his starting position?**

- a) 20 km
- b) 30 km
- c) 50 km
- d) 60 km

Correct Answer: a) 20 km

14. **Two bus tickets from city A to B and three tickets from city A to C cost Rs. 77 but three tickets from city A to B and two tickets from city A to C cost Rs. 73. What are the fares for cities B and C from A ?**

- a) Rs. 4, Rs. 23
- b) Rs. 13, Rs. 17
- c) Rs. 15, Rs. 14
- d) Rs. 17, Rs. 13

Correct Answer: b) Rs. 13, Rs. 17

15. **Ayush was born two years after his father's marriage. His mother is five years younger than his father but 20 years older than Ayush who is 10 years old. At what age did the father get married ?**

- a) 23 years
- b) 25 years
- c) 33 years
- d) 35 years

Correct Answer: a) 23 years

Aptitude/Math Section

1. Two, trains, one from Howrah to Patna and the other from Patna to Howrah, start simultaneously. After they meet, the trains reach their destinations after 9 hours and 16 hours respectively. The ratio of their speeds is:

a) 2 : 3
b) 4 : 3
c) 6 : 7
d) 9 : 16

Correct Answer: b) 4 : 3

2. The length of the bridge, which a train 130 metres long and travelling at 45 km/hr can cross in 30 seconds, is:

a) 200m
b) 225m
c) 245m
d) 250m

Correct Answer: c) 245m

3. A shopkeeper sells an article for \$500, making a profit of 20%. What was the cost price of the article?

a) \$400
b) \$350
c) \$450
d) \$300

Correct Answer: a) \$400

4. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is :

a) $\frac{1}{4}$
b) $\frac{1}{10}$
c) $\frac{7}{15}$
d) $\frac{8}{15}$

Correct Answer: d) $\frac{8}{15}$

5. The sum of the ages of A and B is 50 years. A is 10 years older than B.

What is the age of A?

- a) 30 years
- b) 25 years
- c) 35 years
- d) 40 years

Correct Answer: c) 35 years

6. From a point P on a level ground, the angle of elevation of the top tower is 30° . If the tower is 100 m high, the distance of point P from the foot of the tower is:

- a) 149 m
- b) 156 m
- c) 173 m
- d) 260

Correct Answer: c) 173 m

7. If the cost price of 15 articles is equal to the selling price of 12 articles, what is the profit percent?

- a) 15%
- b) 20%
- c) 25%
- d) 30%

Correct Answer: b) 25%

8. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?

- a) Rs. 1090
- b) Rs. 1160
- c) Rs. 1190
- d) Rs. 1202

Correct Answer: c) Rs. 1190

9. What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?

- a) 1
- b) 14
- c) 21
- d) 20

Correct Answer: d) 20

10. A sum of money is invested at 5% simple interest for 3 years. If the total interest earned is \$150, what is the principal amount?

- a) \$1000
- b) \$1200
- c) \$1500
- d) \$1700

Correct Answer: a) \$1000

11. The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?

- a) 0
- b) 1
- c) 10
- d) 19

Correct Answer: d) 19

12. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

- a) 4
- b) 7
- c) 9
- d) 13

Correct Answer: a) 4



RICR

13. If one-third of one-fourth of a number is 15, then three-tenth of that number is:

- a) 35
- b) 36
- c) 45
- d) 54

Correct Answer: d) 54

14. What is the area of a cylinder with height 5 cm and radius 2 cm?

- a) 87.96 cm²
- b) 85.01 cm²
- c) 91.87 cm²
- d) 105.87 cm²

Correct Answer: a) 87.96 cm²

15. If the cost of 7 oranges is \$14, what is the cost of 15 oranges?

- a) \$25
- b) \$30
- c) \$28
- d) \$35

Correct Answer: c) \$30

Computer Basic Knowledge Section

1. Which of the following types of memory is non-volatile and commonly used for storing firmware?

- a) RAM
- b) ROM
- c) Cache
- d) Virtual Memory

Correct Answer: b) ROM

2. What is the main difference between a process and a thread in operating systems?

- a) Threads are heavier than processes.
- b) Processes share the same memory space, threads do not.
- c) Threads share the same memory space, processes do not.
- d) Threads cannot be executed independently, processes can.

Correct Answer: c) Threads share the same memory space, processes do not.

3. In networking, what does the term "subnet mask" refer to?

- a) A unique identifier for each device on the network.
- b) A method for dividing an IP network into segments.
- c) A protocol for secure communication.
- d) A system for converting domain names to IP addresses.

Correct Answer: b) A method for dividing an IP network into segments.

4. Which of the following is NOT a feature of Object-Oriented Programming (OOP)?

- a) Encapsulation
- b) Polymorphism
- c) Abstraction
- d) Recursion

Correct Answer: d) Recursion

5. **What is the role of the kernel in an operating system?**

- a) Managing system security
- b) Handling I/O operations
- c) Acting as a bridge between hardware and software
- d) Running application programs

Correct Answer: c) Acting as a bridge between hardware and software

6. **What is the maximum length of a filename in Windows?**

- a) 255 characters
- b) 128 characters
- c) 512 characters
- d) 64 characters

Correct Answer: a) 255 characters

7. **In databases, what does "ACID" stand for?**

- a) Atomicity, Consistency, Isolation, Durability
- b) Accuracy, Consistency, Isolation, Dependency
- c) Access, Consistency, Integration, Durability
- d) Availability, Control, Isolation, Dependency

Correct Answer: a) Atomicity, Consistency, Isolation, Durability

8. **What is the full form of 'URL'?**

- a) Universal Resource Locator
- b) Uniform Resource Locator
- c) Uniform Result Locator
- d) Unified Resource Locator

Correct Answer: b) Uniform Resource Locator

9. **What is the primary use of a hard drive in a computer?**

- a) To store data permanently
- b) To process data
- c) To manage temporary memory
- d) To connect peripherals

Correct Answer: a) To store data permanently

10. Which key combination is used to open the Task Manager in Windows?

- a) Ctrl + Alt + Delete
- b) Ctrl + Shift + Esc
- c) Alt + Tab
- d) Ctrl + Esc

Correct Answer: a) Ctrl + Alt + Delete

11. What is the basic unit of data storage in a computer?

- a) Byte
- b) Bit
- c) Nibble
- d) Word

Correct Answer: b) Bit

12. What does the F5 key typically do in a browser?

- a) Opens the help menu
- b) Refreshes the page
- c) Saves the page
- d) Prints the page

Correct Answer: b) Refreshes the page

13. In computer architecture, what is the main purpose of a "cache memory"?

- a) To permanently store data
- b) To speed up access to frequently used data
- c) To increase RAM capacity
- d) To backup system files

Correct Answer: b) To speed up access to frequently used data

14. What is the role of an interrupt in a computer system?

- a) To execute a task repeatedly
- b) To handle errors in program execution
- c) To signal the processor to attend to an urgent task
- d) To allocate memory to processes

Correct Answer: c) To signal the processor to attend to an urgent task

15. What is the primary function of the "Recycle Bin" in Windows?

- a) To permanently delete files
- b) To temporarily store deleted files
- c) To increase disk space
- d) To manage file permissions

Correct Answer: b) To temporarily store deleted files

